Year 13 Further Maths - Differential equations

Similar questions - Integrating factors and rates of change

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A skydiver of mass m falls vertically with speed v m s^{-1} at time t seconds, with $\dfrac{\mathrm{d}v}{\mathrm{d}t} = 10 - 0.5v$ for $0 \leqslant t \leqslant 4$ before the parachute opens, and $\dfrac{\mathrm{d}v}{\mathrm{d}t} = 10 - 2v$ afterwards. Initially $v = 0$.

(a) Solve the first equation using an integrating factor, and find the speed when the parachute opens. **[6]**

(b) State the terminal speed before and after the parachute opens. **[2]**

(c) Find, correct to 2 decimal places, how long after the parachute opens the speed has fallen to 8 m s^{-1}. **[6]**

2 A quantity v satisfies $\dfrac{\mathrm{d}v}{\mathrm{d}t} + \dfrac{2}{t+1}v = 6$ for $t \geqslant 0$, with $v = 1$ when $t = 0$.

(a) Find v in terms of t. **[6]**

(b) Find the mean value of v over $0 \leqslant t \leqslant 3$. **[3]**

(c) Comment on the behaviour of v for large t. **[1]**

3 A tank holds 200 litres of water in which there is initially no salt. Brine of concentration 0.6 g litre^{-1} flows in at 5 litres per minute and the well-stirred mixture flows out at 5 litres per minute. Let Q grams be the mass of salt at time t minutes.

(a) Show that $\dfrac{\mathrm{d}Q}{\mathrm{d}t} = 3 - \dfrac{Q}{40}$. **[3]**

(b) Solve this equation to find Q in terms of t. **[5]**

(c) Find the time at which the concentration in the tank reaches 0.3 g litre^{-1}, and the time at which it reaches 0.5 g litre^{-1}. **[4]**

(d) State the limiting concentration and justify it from the model. **[2]**

Question 1

(a) $v' + 0.5v = 10$; IF $\mathrm{e}^{t/2}$: $\left(\mathrm{e}^{t/2}v\right)' = 10\mathrm{e}^{t/2}$, $v = 20 + c\mathrm{e}^{-t/2}$, $v(0) = 0 \Rightarrow c = -20$. $v = 20(1 - \mathrm{e}^{-t/2})$; at $t = 4$, $v = 20(1 - \mathrm{e}^{-2}) = 17.29\ \mathrm{m\,s^{-1}}$

(b) Before: $20\ \mathrm{m\,s^{-1}}$; after: $5\ \mathrm{m\,s^{-1}}$

(c) With $s = t - 4$: $v = 5 + (17.29 - 5)\mathrm{e}^{-2s}$. $v = 8$: $\mathrm{e}^{-2s} = \frac{3}{12.29}$, $s = \frac{1}{2}\ln\frac{12.29}{3} = 0.71$ s

Question 2

(a) IF $\mathrm{e}^{\int 2/(t+1)\mathrm{d}t} = (t+1)^2$: $\left((t+1)^2 v\right)' = 6(t+1)^2$, so $(t+1)^2 v = 2(t+1)^3 + c$. $v(0) = 1 \Rightarrow c = -1$: $v = 2(t+1) - \dfrac{1}{(t+1)^2}$

(b) $\frac{1}{3}\int_0^3 \left(2(t+1) - (t+1)^{-2}\right)\mathrm{d}t = \frac{1}{3}\left[(t+1)^2 + (t+1)^{-1}\right]_0^3 = \frac{1}{3}(16.25 - 2) = 4.75$

(c) The second term tends to 0 so $v \approx 2(t+1)$ grows linearly

Question 3

(a) Salt in: $5 \times 0.6 = 3$ g/min. Concentration is $Q/200$, so salt out $= 5 \times \frac{Q}{200} = \frac{Q}{40}$

(b) IF $\mathrm{e}^{t/40}$: $\left(\mathrm{e}^{t/40}Q\right)' = 3\mathrm{e}^{t/40}$, $Q = 120 + c\mathrm{e}^{-t/40}$, $Q(0) = 0$: $Q = 120\left(1 - \mathrm{e}^{-t/40}\right)$

(c) Concentration 0.3: $Q = 60$, $\mathrm{e}^{-t/40} = \frac{1}{2}$, $t = 40\ln 2 = 27.7$ min. Concentration 0.5: $Q = 100$, $\mathrm{e}^{-t/40} = \frac{1}{6}$, $t = 40\ln 6 = 71.7$ min

(d) $Q \to 120$ so the concentration tends to 0.6 $\mathrm{g\,litre^{-1}}$, the concentration of the incoming brine